

Google Cloud Security Engineering Lab

Part 1: Identity and Access Management (IAM) Security Architecture

Author

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Domain

Cloud Security Engineering

Platform

Google Cloud Platform (GCP)

Executive Summary

This project demonstrates the implementation of Identity and Access Management (IAM) controls within a Google Cloud environment. The objective was to understand how IAM roles, custom roles, service accounts, and access control policies can be used to secure cloud infrastructure and enforce the Principle of Least Privilege.

The project consists of multiple users, projects, IAM role assignments, access validation tests, and service account implementations. Security testing was performed to verify authorization controls and resource governance mechanisms.

Project Objective

The primary objectives of this project were:

- Understand Google Cloud IAM architecture
- Implement Role-Based Access Control (RBAC)
- Configure predefined IAM roles
- Create custom IAM roles
- Configure service accounts
- Demonstrate least privilege access
- Validate permission enforcement
- Secure Compute Engine resources

Environment Setup

Cloud Platform

Google Cloud Platform (GCP)

Resources Used

- IAM
- Compute Engine
- Service Accounts
- Cloud Shell
- gcloud CLI
- Cloud Audit Logs

Projects

Project 1

- centos-clean
- lab-1

Project 2

- lab-2
- lab-3
- lab-4

Architecture Overview

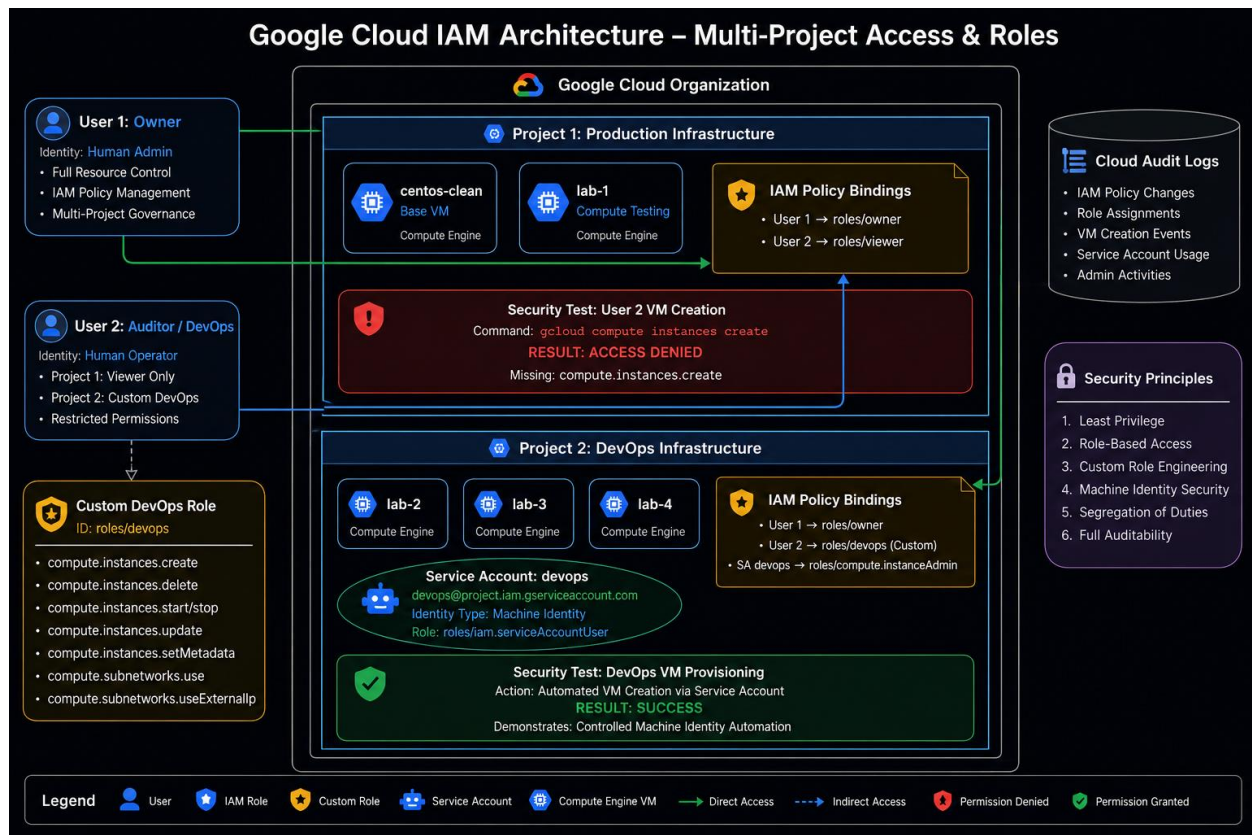


Figure 1:

Google Cloud IAM Security Architecture showing Owner, Viewer, DevOps Role, Service Account, and Compute Engine resources.

The architecture demonstrates how users receive different permissions based on assigned IAM roles and how service accounts provide secure machine identities for automation.

Scenario 1: Owner Access Configuration

User 1 was assigned the Owner role for both projects.

Responsibilities:

- Resource Administration
- IAM Management
- Service Account Management
- Compute Engine Administration

Permissions:

- Create Resources
- Modify Resources
- Delete Resources
- Assign IAM Policies

```
You are now logged in as [student-03-9cab74ff5ac1@qwiklabs.net].
Your current project is [None]. You can change this setting by running:
$ gcloud config set project PROJECT_ID
student_03_9cab74ff5ac1@cloudshell:~$ gcloud config set project qwiklabs-gcp-04-c2d55474e287
[environment: untagged] Read more to tag: g.co/cloud/project-env-tag.
Updated property [core/project].
student_03_9cab74ff5ac1@cloudshell:~ (qwiklabs-gcp-04-c2d55474e287) $ gcloud config set compute/region us-east4
gcloud config set compute/zone us-east4-b
Updated property [compute/region].
Updated property [compute/zone].
student_03_9cab74ff5ac1@cloudshell:~ (qwiklabs-gcp-04-c2d55474e287) $ gcloud compute instances create lab-1 --zone us-east4-b --machine-type=e2-standard-2
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-c2d55474e287/zones/us-east4-b/instances/lab-1].
NAME: lab-1
ZONE: us-east4-b
MACHINE_TYPE: e2-standard-2
PREEMPTIBLE:
INTERNAL_IP: 10.150.0.3
EXTERNAL_IP: 34.11.23.32
STATUS: RUNNING
student_03_9cab74ff5ac1@cloudshell:~ (qwiklabs-gcp-04-c2d55474e287) $
```

Figure 2:

Owner account successfully managing cloud resources.

Result:

Owner role provides full administrative control over both projects.

Scenario 2: Viewer Access Restriction

User 2 was assigned the Viewer role.

The Viewer role allows users to inspect resources but does not allow infrastructure modifications.

Permissions:

- View Resources
- View Configurations
- Read-only Access

Restrictions:

- Cannot Create VM
- Cannot Modify VM
- Cannot Delete Resources

```
[student-03-9cab74ff5ac1@centos-clean ~]$ echo "export USERID2=student-04-e5196belc7fa@qwiklabs.net" >> ~/.bashrc
[student-03-9cab74ff5ac1@centos-clean ~]$ . ~/.bashrc
gcloud projects add-iam-policy-binding $PROJECTID2 --member user:$USERID2 --role=roles/viewer
Updated IAM policy for project [qwiklabs-gcp-03-b216cb8cdcfe].
bindings:
- members:
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcfe@qwiklabs-gcp-03-b216cb8cdcfe.iam.gserviceaccount.com
  role: roles/bigquery.admin
- members:
  - serviceAccount:204460198661@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-204460198661@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:204460198661@cloudservices.gserviceaccount.com
  role: roles/compute.instanceGroupManagerServiceAgent
- members:
  - serviceAccount:service-204460198661@compute-system.iam.gserviceaccount.com
  role: roles/compute.serviceAgent
- members:
  - serviceAccount:204460198661-compute@developer.gserviceaccount.com
  - serviceAccount:204460198661@cloudservices.gserviceaccount.com
  role: roles/editor
- members:
  - user:student-03-9cab74ff5ac1@qwiklabs.net
  role: roles/iam.serviceAccountAdmin
- members:
  - serviceAccount:admiral@qwiklabs-services-prod.iam.gserviceaccount.com
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcfe@qwiklabs-gcp-03-b216cb8cdcfe.iam.gserviceaccount.com
  - user:student-03-9cab74ff5ac1@qwiklabs.net
  role: roles/owner
- members:
  - user:student-03-9cab74ff5ac1@qwiklabs.net
  role: roles/resourcemanager.projectIamAdmin
- members:
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcfe@qwiklabs-gcp-03-b216cb8cdcfe.iam.gserviceaccount.com
  role: roles/storage.admin
- members:
  - user:student-03-9cab74ff5ac1@qwiklabs.net
  - user:student-04-e5196belc7fa@qwiklabs.net
  role: roles/viewer
etag: BwZU22i7ibo=
version: 1
[student-03-9cab74ff5ac1@centos-clean ~]$
```

Figure 3:
Viewer role assignment in IAM.

Scenario 3: Authorization Failure Test

To validate IAM controls, User 2 attempted to create a new virtual machine.

Command Executed:

gcloud compute instances create lab-2

Result:

ACCESS DENIED

Reason:

Missing Permission

Compute.instances.create

```
[student-03-9cab74ff5ac1@centos-clean ~]$ gcloud compute instances list
NAME          ZONE          MACHINE_TYPE  PREEMPTIBLE  INTERNAL_IP  EXTERNAL_IP  STATUS
centos-clean  us-east4-b    e2-standard-2  10.150.0.2   34.85.182.98  RUNNING
lab-1         us-east4-b    e2-standard-2  10.150.0.4   34.11.23.32   RUNNING
[student-03-9cab74ff5ac1@centos-clean ~]$ gcloud compute instances create lab-2 --zone us-east1-c --machine-type=e2-standard-2
ERROR: (gcloud.compute.instances.create) Could not fetch resource:
- Required 'compute.instances.create' permission for 'projects/qwiklabs-gcp-04-c2d55474e287/zones/us-east1-c/instances/lab-2'

[student-03-9cab74ff5ac1@centos-clean ~]$
```

Figure 4:

VM creation failed because Viewer permissions were insufficient.

Security Analysis:

This demonstrates the Principle of Least Privilege, where users receive only the permissions required for their responsibilities.

Scenario 4: Custom DevOps Role Creation

A custom IAM role named DevOps was created to provide operational permissions without granting full Owner privileges.

Permissions Included:

- compute.instances.create
- compute.instances.delete
- compute.instances.start
- compute.instances.stop
- compute.instances.update
- Compute.instances.setMetadata

```
[student-03-9cab74ff5a1@centos-clean ~]$ gcloud projects add-iam-policy-binding $PROJECTID2 --member user:$USERID2 --role=roles/iam.serviceAccountUser
Updated IAM policy for project [qwiklabs-gcp-03-b216cb8cdcf].
bindings:
- members:
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcf@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  role: roles/bigquery.admin
- members:
  - serviceAccount:204460198661@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-204460198661@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:204460198661@cloudservices.gserviceaccount.com
  role: roles/compute.instanceGroupManagerServiceAgent
- members:
  - serviceAccount:service-204460198661@compute-system.iam.gserviceaccount.com
  role: roles/compute.serviceAgent
- members:
  - serviceAccount:204460198661-compute@developer.gserviceaccount.com
  - serviceAccount:204460198661@cloudservices.gserviceaccount.com
  role: roles/editor
- members:
  - user:student-03-9cab74ff5a1@qwiklabs.net
  role: roles/iam.serviceAccountAdmin
- members:
  - user:student-04-e5196belc7fa@qwiklabs.net
  role: roles/iam.serviceAccountUser
- members:
  - serviceAccount:admiral@qwiklabs-services-prod.iam.gserviceaccount.com
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcf@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  - user:student-03-9cab74ff5a1@qwiklabs.net
  role: roles/owner
- members:
  - user:student-03-9cab74ff5a1@qwiklabs.net
  role: roles/resourcemanager.projectIamAdmin
- members:
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcf@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  role: roles/storage.admin
- members:
  - user:student-03-9cab74ff5a1@qwiklabs.net
  - user:student-04-e5196belc7fa@qwiklabs.net
  role: roles/viewer
etag: BwZU2Z7-II=
version: 1
[student-03-9cab74ff5a1@centos-clean ~]$ gcloud projects add-iam-policy-binding $PROJECTID2 --member user:$USERID2 --role=projects/$PROJECTID2/roles/devops
Updated IAM policy for project [qwiklabs-gcp-03-b216cb8cdcf].
bindings:
- members:
  - user:student-04-e5196belc7fa@qwiklabs.net
  role: projects/qwiklabs-gcp-03-b216cb8cdcf/roles/devops
- members:
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcf@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  role: roles/bigquery.admin
- members:
  - serviceAccount:204460198661@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-204460198661@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:204460198661@cloudservices.gserviceaccount.com
  role: roles/compute.instanceGroupManagerServiceAgent
- members:
  - serviceAccount:service-204460198661@compute-system.iam.gserviceaccount.com
  role: roles/compute.serviceAgent
- members:
  - serviceAccount:204460198661-compute@developer.gserviceaccount.com
  - serviceAccount:204460198661@cloudservices.gserviceaccount.com
  role: roles/editor
- members:
  - user:student-03-9cab74ff5a1@qwiklabs.net
  role: roles/iam.serviceAccountAdmin
```

Figure 5:
Custom DevOps role configuration.

Security Analysis:

Custom roles provide granular access control and reduce the risk associated with excessive permissions.

Scenario 5: IAM Policy Binding

The DevOps role was assigned to User 2 through IAM policy bindings.

Assigned Roles:

- Viewer

- DevOps
- Service Account User

```
[student-03-9cab74ff5a1@centos-clean ~]$ gcloud projects add-iam-policy-binding $PROJECTID2 --member serviceAccount:$SA --role=roles/compute.instanceAdmin
Updated IAM policy for project [qwiklabs-gcp-03-b216cb8cdcf].
bindings:
- members:
  - user:student-04-e5196belc7fa@qwiklabs.net
    role: projects/qwiklabs-gcp-03-b216cb8cdcf/roles/devops
- members:
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcf@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
    role: roles/bigquery.admin
- members:
  - serviceAccount:204460198661@cloudbuild.gserviceaccount.com
    role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-204460198661@gcp-sa-cloudbuild.iam.gserviceaccount.com
    role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:devops@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
    role: roles/compute.instanceAdmin
- members:
  - serviceAccount:204460198661@cloudservices.gserviceaccount.com
    role: roles/compute.instanceGroupManagerServiceAgent
- members:
  - serviceAccount:service-204460198661@compute-system.iam.gserviceaccount.com
    role: roles/compute.serviceAgent
- members:
  - serviceAccount:204460198661-compute@developer.gserviceaccount.com
  - serviceAccount:204460198661@cloudservices.gserviceaccount.com
    role: roles/editor
- members:
  - user:student-03-9cab74ff5a1@qwiklabs.net
    role: roles/iam.serviceAccountAdmin
- members:
  - serviceAccount:devops@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  - user:student-04-e5196belc7fa@qwiklabs.net
    role: roles/iam.serviceAccountUser
- members:
  - serviceAccount:admiral@qwiklabs-services-prod.iam.gserviceaccount.com
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcf@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  - user:student-03-9cab74ff5a1@qwiklabs.net
    role: roles/owner
- members:
  - user:student-03-9cab74ff5a1@qwiklabs.net
    role: roles/resourceManager.projectIamAdmin
- members:
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcf@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
    role: roles/storage.admin
- members:
  - user:student-03-9cab74ff5a1@qwiklabs.net
  - user:student-04-e5196belc7fa@qwiklabs.net
    role: roles/viewer
etag: BwZU2afSGzE=
version: 1
```

Figure 6:

User 2 assigned additional permissions.

Result:

User 2 successfully gained operational access while maintaining controlled permissions.

Scenario 6: Successful VM Creation

After receiving the DevOps role, User 2 attempted VM creation again.

Result:

SUCCESSFUL

Resources Created:

lab-2


```
[student-03-9cab74ff5ac1@centos-clean ~]$ gcloud config configurations activate user2
Activated [user2].
[student-03-9cab74ff5ac1@centos-clean ~]$ gcloud compute instances create lab-2 --zone us-east1-c --machine-type=e2-standard-2
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-03-b216cb8cdcf/zones/us-east1-c/instances/lab-2].
NAME      ZONE      MACHINE_TYPE  PREEMPTIBLE  INTERNAL_IP  EXTERNAL_IP  STATUS
lab-2     us-east1-c  e2-standard-2  10.142.0.2   34.23.14.201  RUNNING
[student-03-9cab74ff5ac1@centos-clean ~]$
```

Figure 7:

Successful virtual machine deployment after role assignment.

Security Analysis:

RBAC successfully enabled controlled infrastructure management.

Scenario 7: Service Account Configuration

A service account named devops was created to support secure automation.

Purpose:

- Infrastructure Automation
 - Secure Authentication
- Non-Human Identity

```
[student-03-9cab74ff5ac1@centos-clean ~]$ gcloud projects add-iam-policy-binding $PROJECTID2 --member serviceAccount:$SA --role=roles/compute.instanceAdmin
Updated IAM policy for project [qwiklabs-gcp-03-b216cb8cdcf].
bindings:
- members:
  - user:student-04-e5196b6c7fa@qwiklabs.net
  role: projects/qwiklabs-gcp-03-b216cb8cdcf/roles/devops
- members:
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcf@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  role: roles/bigquery.admin
- members:
  - serviceAccount:204460198661@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-204460198661@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:devops@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  role: roles/compute.instanceAdmin
- members:
  - serviceAccount:204460198661@cloudservices.gserviceaccount.com
  role: roles/compute.instanceGroupManagerServiceAgent
- members:
  - serviceAccount:service-204460198661@compute-system.iam.gserviceaccount.com
  role: roles/compute.serviceAgent
- members:
  - serviceAccount:204460198661-compute@developer.gserviceaccount.com
  - serviceAccount:204460198661@cloudservices.gserviceaccount.com
  role: roles/editor
- members:
  - user:student-03-9cab74ff5ac1@qwiklabs.net
  role: roles/iam.serviceAccountAdmin
- members:
  - serviceAccount:devops@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  - user:student-04-e5196b6c7fa@qwiklabs.net
  role: roles/iam.serviceAccountUser
- members:
  - serviceAccount:admiral@qwiklabs-services-prod.iam.gserviceaccount.com
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcf@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  - user:student-03-9cab74ff5ac1@qwiklabs.net
  role: roles/owner
- members:
  - user:student-03-9cab74ff5ac1@qwiklabs.net
  role: roles/resourcemanager.projectIamAdmin
- members:
  - serviceAccount:qwiklabs-gcp-03-b216cb8cdcf@qwiklabs-gcp-03-b216cb8cdcf.iam.gserviceaccount.com
  role: roles/storage.admin
- members:
  - user:student-03-9cab74ff5ac1@qwiklabs.net
  - user:student-04-e5196b6c7fa@qwiklabs.net
  role: roles/viewer
etag: BwZU2afSGzE=
version: 1
```

Figure 8:

Service account configuration.

Benefits:

- Reduced dependency on user credentials
- Improved security posture
- Auditable infrastructure operations

Scenario 8: Service Account Authorization

The service account received:

- roles/iam.serviceAccountUser
- roles/compute.instanceAdmin

Permissions:

- Create VMs
- Modify VMs
- Delete VMs
- Manage Compute Resources

Figure 9:

Service account role configuration.

Scenario 9: Infrastructure Automation Validation

Using the service account, additional Compute Engine resources were created.

Resources:

- lab-3
- lab-4

```
Your active configuration is: [default]
[environment: untagged] Read more to tag: g.co/cloud/project-env-tag.
[student-03-9cab74ff5acl@centos-clean ~]$ gcloud compute instances create lab-4 --zone us-east1-c --machine-type=e2-standard-2
Created [https://www.googleapis.com/compute/v1/projects/gwiklabs-gcp-03-b216cb8cdcfe/zones/us-east1-c/instances/lab-4].
NAME      ZONE      MACHINE_TYPE  PREEMPTIBLE  INTERNAL_IP  EXTERNAL_IP  STATUS
lab-4     us-east1-c  e2-standard-2  10.142.0.4   34.138.252.49  RUNNING
[student-03-9cab74ff5acl@centos-clean ~]$ gcloud compute instances list
NAME      ZONE      MACHINE_TYPE  PREEMPTIBLE  INTERNAL_IP  EXTERNAL_IP  STATUS
lab-2     us-east1-c  e2-standard-2  10.142.0.2   34.23.14.201  RUNNING
lab-3     us-east1-c  e2-standard-2  10.142.0.3   34.24.123.250  RUNNING
lab-4     us-east1-c  e2-standard-2  10.142.0.4   34.138.252.49  RUNNING
[student-03-9cab74ff5acl@centos-clean ~]$
```

Figure 10:

Infrastructure managed through service account permissions.

Result:

Service account successfully performed administrative operations without using a personal user account.

Security Principles Demonstrated

- Identity and Access Management (IAM)
- Principle of Least Privilege
- Role-Based Access Control (RBAC)
- Service Account Security
- Access Governance
- Segregation of Duties
- Accountability
- Auditability

Key Findings

Finding	Result
Viewer cannot create VMs	Validated
IAM policies enforce permissions	Validated
Custom DevOps role works correctly	Validated
Service accounts support automation	Validated
RBAC successfully controls access	Validated

Skills Demonstrated

- Google Cloud IAM
- Cloud Security
- Role-Based Access Control
- Service Account Management
- Compute Engine Administration
- Access Governance
- Security Documentation
- Google Cloud CLI

Conclusion

This project successfully demonstrated how Google Cloud IAM can be used to secure cloud environments through role-based access control, custom role creation, service account management, and least-privilege principles.

The implementation validated real-world cloud security concepts used by cloud administrators, security engineers, and DevOps teams in enterprise environments.